

Solving Inequalities in Real-World ContextsWorksheet 7A.01.10 **ANSWER KEY**

Quinn solved the inequality $2.5 \geq (c - 3)/4$ and made an error. Her work is shown. Use it for questions 1 through 3.

Quinn's work

$$2.5 \geq (c - 3) / 4$$

$$2.5 + 3 \geq (c - 3 + 3) / 4$$

$$5.5 \geq c / 4$$

$$22 \geq c$$

1. What error did Quinn make?

She added 3 to the left side but only to the numerator on the right.

Adding 3 inside the numerator adds $3/4$ to the value of that side, not 3. The two sides were not changed by the same amount, so the statement stopped being equivalent.

2. Write a brief note to Quinn explaining how to avoid this in the future. Say what she did correctly as well.

Sample: "Your idea was right — you wanted to undo the $- 3$. What went wrong is where you put the $+ 3$. Because the whole expression $c - 3$ is divided by 4, adding 3 to the top only adds $3/4$ to that side. Clear the division first: multiply BOTH sides by 4, then add 3 to both sides. Your arithmetic was clean the whole way, and you kept the symbol pointing the right way, which is the part most people get wrong."

Answers will vary, but a complete note names the specific error, gives a working alternative, and identifies something correct. Quinn never needed to reverse the symbol and she did not, which is worth telling her.

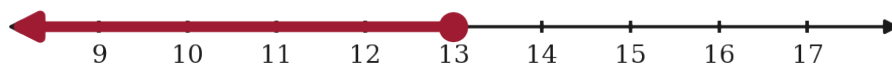
3. Solve Quinn's inequality correctly and represent the solution on the number line.

$$2.5 \geq (c - 3) / 4$$

$$10 \geq c - 3 \quad (\text{multiply both sides by 4})$$

$$13 \geq c, \text{ so } c \leq 13$$

Check: $c = 13$ gives $(10)/4 = 2.5$, and $2.5 \geq 2.5 \checkmark$; $c = 20$ gives $17/4 = 4.25$, and $2.5 \geq 4.25 \times$



Closed circle at 13, shaded left. Nothing reverses — 4 is positive.

For questions 4 through 7, solve the inequality and represent the solution on the number line.

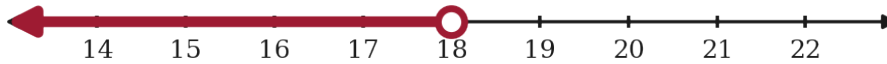
4. $-\frac{b}{4} + 5 > \frac{1}{2}$

$-\frac{b}{4} > -4.5$ (subtract 5 from both sides)

$b < 18$ (multiply both sides by -4 and REVERSE)

Check: $b = 0$ gives $5 > 0.5$ ✓; $b = 20$ gives $-5 + 5 = 0 > 0.5$ ✗

Solution. $b < 18$



Multiplying by -4 reverses the symbol once.

5. $8.8 \geq 0.8(-17.3 + 2.9x)$

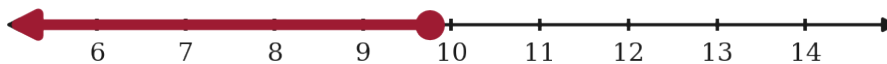
$11 \geq -17.3 + 2.9x$ (divide both sides by 0.8 — no reversal)

$28.3 \geq 2.9x$

$x \leq 283/29 \approx 9.76$

Check: $x = 9$ gives $0.8(8.8) = 7.04$ and $8.8 \geq 7.04$ ✓; $x = 11$ gives $0.8(14.6) = 11.68$ and $8.8 \geq 11.68$ ✗

Solution. $x \leq 9.76$ (about)



The boundary is not a whole number. Put the closed circle between 9 and 10, closer to 10, and shade left. Saying “between 9 and 10” in words is part of the answer — an exact tick mark does not exist for this one.

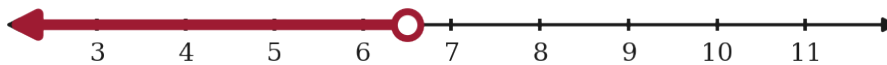
6. $1.5m - 10 < -\frac{1}{4}$

$1.5m < 9.75$ (add 10 to both sides; $-\frac{1}{4} = -0.25$)

$m < 6.5$

Check: $m = 6$ gives $9 - 10 = -1$ and $-1 < -0.25$ ✓; $m = 7$ gives $10.5 - 10 = 0.5$ and $0.5 < -0.25$ ✗

Solution. $m < 6.5$



6.5 sits exactly halfway between the 6 and 7 tick points. Open circle there, shade left.

7. $-\frac{2}{3}(-3 - 12a) \leq -6$

$-3 - 12a \geq 9$ (multiply both sides by $-\frac{3}{2}$ and REVERSE)

$-12a \geq 12$

$a \leq -1$ (divide both sides by -12 and REVERSE again)

Check: $a = -1$ gives $-\frac{2}{3}(9) = -6$ and $-6 \leq -6$ ✓; $a = 0$ gives $-\frac{2}{3}(-3) = 2$ and $2 \leq -6$ ✗

Solution. $a \leq -1$



Two reversals here. Count them as you go — the answer is only backwards if you make an odd number of mistakes.

Mary is buying donuts and juice for her son's third birthday party at school. She has at most \$25 to spend. She bought 10 donuts at \$1.25 each, and each case of juice costs \$3.49. Mary wrote $10(1.25) + 3.49c \leq 25$, where c is the number of cases of juice.

8. Why did Mary use $\leq$ rather than $<$, $>$ or $\geq$?

“At most \$25” means she may spend \$25 or any amount less than that, so the total is less than OR equal to 25.

$>$ and $\geq$ point the wrong way — they would let her spend more than she has. A strict $<$ would rule out spending exactly \$25, which she is allowed to do.

9. Solve the inequality.

$12.50 + 3.49c \leq 25$

$3.49c \leq 12.50$

$c \leq 3.58\dots$ ($12.50 \div 3.49 = 3.5816\dots$)

Mary can buy at most 3 cases of juice.

Round DOWN. Juice comes in whole cases and 3.58 is a ceiling, so 3 is the largest number that fits. 3 cases cost $12.50 + 10.47 = \$22.97$.

10. Will Mary be able to purchase 4 cases of juice? Justify your answer mathematically.

No. $10(1.25) + 3.49(4) = 12.50 + 13.96 = \26.46

\$26.46 is more than \$25, so 4 cases does not satisfy the inequality. She is short by \$1.46.